IPCC Tier 2 Cattle GHG Uncertainty Calculator

User Guide

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1 Overview

1.1 What this tool does

Cattle are a major source of methane. In many countries their emissions are still growing. Countries can estimate these emissions more accurately with the IPCC Tier 2 method, which uses national data on the animals, their feed and their manure. The Paris Agreement transparency rules encourage countries to use these higher tier methods and to keep improving their inventories.

Every estimate has an uncertainty: the true value may be somewhat higher or lower than the number reported. The IPCC asks countries to say how large this uncertainty is. This is good practice whenever a higher tier method is used. It is also useful in itself: once you know which inputs drive the uncertainty, you know where better data would improve the inventory most.

Many countries cannot do this analysis today. The Tier 2 equations are complex, so the uncertainty has to be estimated with a Monte Carlo simulation. Inventory compilers and livestock experts rarely have the statistical training for this. The commercial software that does it is expensive, and the free options do not show which inputs drive the uncertainty.

This tool fills that gap. It is free, it runs in a web browser, and it does the Monte Carlo simulation for you. In short:

1. You give the tool your Tier 2 inputs (animal numbers, weights, milk yields, emission factors and so on), each with an uncertainty range.
2. The tool draws thousands of possible values for every input, within the ranges you gave, and runs the full IPCC Tier 2 calculation for each set of draws.
3. It reports the range in which your emissions most likely lie (the 95% confidence interval), shows which inputs matter most, and gives you the numbers in the format of IPCC Table 3.3 for your inventory report.

The method is IPCC Approach 2 (Monte Carlo), as described in the 2006 IPCC Guidelines, Volume 1, Chapter 3.

1.2 Who this tool is for

This tool is for national inventory practitioners who:

- are already doing, or have recently completed, a Tier 2 cattle GHG inventory;
- have at least rough estimates of the uncertainty of their main inputs;
- need to report an uncertainty range in their national inventory report.

You need to know the IPCC Tier 2 method and some basic statistics. You do not need any experience with Monte Carlo simulation. It helps to know what a probability distribution is, and this guide explains the terms as they come up.

2 Preparing your data

2.1 What you need before you start

Before opening the tool, you need the right data to run it. Four kinds of input are needed.

1. **Activity data.** For each animal sub-category: the number of animals (N), the body weight (BW), the daily weight gain (WG), the milk yield (for cows), the milk fat content, the digestibility of the feed (DE), and the other Tier 2 inputs your inventory uses.
2. **Uncertainty ranges.** For each input, an estimate of how uncertain it is. This is usually given in one of two ways:
   - a percentage (for example ±20%, meaning the true value is very likely within 20% of the number you give), or
   - a lower and an upper value.
3. **Manure management fractions.** The share of manure handled in each management system (for example pasture, solid storage, liquid storage).
4. **Emission factors and other coefficients.** If your inventory uses country-specific emission factors (for example EF3 or EF4) or country-specific values for other coefficients (for example C, Ca, Cp or Frac_GasMS), you need those values with their uncertainty. If your inventory uses the IPCC default values, the tool fills them in for you.

2.2 AI-assisted data preparation

Your data will probably not be in the format the tool needs. The tool has an AI translator that converts it for you. It is a small chat panel on the Resources tab. You upload your file, the AI reads it and asks you a few questions, and it produces a filled template that you can download and load into the tool.

The translator works in three steps: it explores your file, it asks you to clarify what is unclear, and it writes the template. The AI translator guide explains each step with screenshots.

The translator maps your columns to the template. It does not check your numbers. Always do two things afterwards: compare the main numbers (animal numbers, body weights, milk yields) with your original file, and look at the QA/QC tab after you load the template. Any value the translator took from an IPCC default is marked there.

3 Loading your data (Tab 1: Data Input)

When you open Tab 1, nothing is loaded yet. First choose whether your analysis is for a single year or for a trend over several years. Then choose your data.

3.1 Using the example datasets

Two example datasets let you try the tool without your own data:

- **Country X (hypothetical dairy)**: a fictional highland dairy system, 12 parameters, dairy cows, with a made-up 5-year time series.
- **Country Y (hypothetical pastoral)**: a fictional semi-arid pastoral beef system, 11 parameters, non-dairy breeding cows, with a made-up 5-year time series.

Pick one in the "Country / Example Data" list and confirm. The data loads at once. You can then go on to Tab 2.

3.2 The parameter table

Once data is loaded, a table appears on the right. It lists every parameter with its mean value, its distribution, its uncertainty percentage, its lower and upper bounds and its type. The table is read-only: the values come from your file. To change a value, edit your Excel file and upload it again. You cannot edit data inside the tool. This is on purpose: always correct your source file first, so that the file and the results stay consistent.

3.3 Uploading your own data

Follow these steps if you want to fill in the template yourself, without the AI translator.

Step 1: Choose an IPCC Guidelines version. In the "Country / Example Data" list, pick "Custom Upload". Then pick the IPCC version your inventory follows:

- **IPCC 2006**: the eight original manure management systems. Choose this if your inventory follows the 2006 Guidelines.
- **IPCC 2019 Refinement**: four more manure management systems (anaerobic digester, aerobic treatment, covered or compacted solid storage, burned for fuel), with the updated emission factors and nitrogen excretion equations. Choose this if your inventory follows the 2019 Refinement.

The template you download only offers the manure systems that exist in the version you picked.

Step 2: Download and fill in the template.

1. Click "Download blank template" to get the Excel workbook.
2. Fill in the **Parameters** sheet: one row per parameter and per animal sub-category.
3. Fill in the **Manure_Management** sheet if you include the manure N_2O sources.

4. Optional: fill in the **Parameter_TimeSeries** sheet if you want the tool to compute correlations from your historical data.
5. Save the file as .xlsx and upload it with the file picker.

4 Reviewing data quality (Tab 2: QA/QC)

4.1 What the QA/QC checks do

The QA/QC tab checks every parameter you loaded and tells you if something looks wrong. Each parameter gets one of five statuses. The tiles at the top count them; click a tile to see only those parameters. Open a row to read each finding and its message.

- **Pass** (green). The value, its unit and its bounds are fine. Nothing to do.
- **Info** (blue). A note, nothing to correct. Example: a parameter whose lower and upper bounds equal its mean is treated as a fixed value. The simulation can run.
- **Warn** (amber). The value is unusual, or something is missing that the tool can work around. Examples: a body weight far from the IPCC default for that animal class; a digestibility outside its usual range; a row without any uncertainty. The simulation can run. Check the value. If it is correct for your country, keep it and note the source in your inventory report. If it is a mistake, correct it in your file.
- **Fail** (red). The value would break or distort the calculation. Examples: bounds that do not contain the mean; a percentage entered as a fraction (DE = 0.65 instead of 65); a fraction entered as a percentage (pct_pregnant = 52 instead of 0.52). The simulation cannot run until every Fail is corrected.
- **Missing** (grey). The parameter was not in your file. The tool filled it with the IPCC default and tells you which value it used. The simulation can run. Check that the default makes sense for your country. If you have a national value, add it to your file.

Make every correction in your Excel file and upload it again on Tab 1. The simulation then uses the values in your file, and the reports document them as such.

5 Correlations (Tab 3: Correlations)

For a worked comparison of the four correlation modes on the two example datasets (how each mode changes the 95% confidence interval, the structural-defaults matrix with its sources, and the common mistakes), click the green "Find out more" link at the top of the Activity Data Correlations card. It opens a topic page in a new browser tab.

5.1 Why correlations matter

Two different things are called "correlation" on this page. They matter to very different degrees.

Between parameters, within one group. Some inputs are linked. For example, heavier animals tend to give more milk. If the tool ignores such a link and draws the two values independently, it understates the uncertainty of the total. For most Tier 2 inventories this effect is small, a few percentage points, and ignoring it is an acceptable simplification. The IPCC recommends including known correlations where they exist.

Between groups, for the same parameter. This one is not small. If several of your groups use the same number for a coefficient, because it comes from one IPCC default or one regional study, that is a single estimate used many times. If it is wrong, every group is wrong in the same direction at once, and nothing cancels out. The tool therefore draws such a value once per iteration for all the groups that use it. Groups with their own value are drawn separately. Nothing needs to be set: the tool reads the values. In the tornado chart a shared value appears as one bar, for example "DE = 52.8 (extensive beef, 6 groups)".

Why this matters: if each group drew its own value instead, the errors would partly cancel and the reported uncertainty would shrink the more finely the inventory is split. On the Zambia inventory (32 groups) the 95% margin of error is 32.5% with shared draws and 23.7% with separate draws; cutting the same herd into 128 groups leaves the first at 30% and takes the second down to 11.5%, with no change in the data. The shared figure is the right one, and it is what the tool reports. The IPCC 2019 Refinement (Volume 1, Chapter 3) makes the same point: sub-categories "can be highly correlated, because either the AD are derived from the same source or the EFs have parameters in common".

5.2 Quick guide: which option do I pick?

- You have five or more years of national data in your Parameter_TimeSeries sheet: pick **From template (auto, time-series)**.
- You have no time series, but you want the known biological links (weight, milk, digestibility): pick **Structural defaults**.
- You have your own correlation matrix from a study or from expert judgement: pick **Advanced: manual entry**.
- None of these: keep **No correlations**. This is the default. It is simple to explain and defend.

An option you cannot use is greyed out, with a short note saying why. For example, "From template"

is greyed out when your Parameter_TimeSeries sheet is empty, and "Advanced: manual entry" needs a CSV file first. If a mode without data is somehow selected, the Run button is blocked with a clear message.

5.3 What change in the result should I expect?

The central estimate (the mean emissions) stays the same whichever mode you pick. What changes is the width of the 95% uncertainty interval. As a rule of thumb, for a typical single-year cattle inventory:

- **No correlations**: the baseline. Every parameter is drawn independently.
- **Structural defaults**: the interval changes by about one percentage point or less on the examples (Country X 25.3% to 25.2%; Country Y 36.5% to 37.5%). The effect is small because the preset links only 7 pairs out of about 78 possible pairs, with moderate values.
- **From template (auto, time-series)**: depends on your data. A rich series where several parameters really move together can have an effect similar to the structural defaults or larger. A sparse or mostly flat series has almost no effect.
- **Advanced: manual entry**: depends entirely on what you upload. Strong correlations on parameters with wide uncertainty can move the interval by 20% or more, in either direction.

If you select correlations and the result barely changes, that is usually expected, not a bug. The IPCC calculation multiplies about 13 parameters together, which dilutes the effect of any single pair. The Methodology document explains the mathematics.

Before you run the simulation, Tab 4 (Simulate & Results) lets you tick "Compare with/without correlations". The run is then done twice, with and without your correlations, so you can see their effect directly.

5.4 Automatic correlations from time-series data

If your upload has a Parameter_TimeSeries sheet, the tool can compute the correlations from your own history:

1. On Tab 3, select **From template (auto, time-series)**.
2. The tool computes the rank correlation between the year-to-year changes of each parameter. Using the changes, and not the raw values, removes the common long-term growth that would otherwise make everything look correlated. The matrix is applied in the simulation with the Iman-Conover method, the one the IPCC cites for this purpose.
3. Look at the matrix on the right of the page and check that the correlations make sense.

5.5 Manual correlation matrix

If you prefer to enter the correlations yourself:

1. Select **Advanced: manual entry**.

2. Upload a square CSV matrix of correlation coefficients. Each value is between -1 and +1, and the diagonal is 1.
3. The tool checks that the matrix is valid for simulation. If it is not, it adjusts it to the nearest valid matrix and warns you when any value moves by more than 0.01.

Two CSV templates appear in the panel, so you never have to build the matrix from scratch:

- **Download blank matrix template**: all parameter names in place, the diagonal set to 1, everything else 0. Open it in Excel, fill in the pairs you want, save as CSV, and upload it.
- **Download matrix with example values**: the same matrix pre-filled with the seven structural-defaults pairs. Use it as a starting point if you only want to change one or two pairs.

The templates already carry the exact parameter names (N, BW, MW, WG, Milk, Fat, pct_pregnant, DE, Cfi, Ca, C, Cp, hours, CP, Ym, Bo, ASH, UE, EF3_PRP, EF4, EF5, Frac_GASM_PRP, Frac_LEACH_PRP, MilkPR, Tw). Do not rename rows or columns. A name the tool does not recognise is ignored without an error message, and the correlation simply does not take effect.

If your matrix leaves out a parameter, every pair with that parameter is treated as independent. The same happens with a misspelled name. The simulation still runs and uses the correlations you did give. This is intentional, so the easiest way to be sure your matrix is used as you expect is to start from the downloadable templates.

5.6 Structural defaults (literature-based)

The built-in preset applies seven pairs. The direction of each link comes from the IPCC equations and from published studies. The size was set by the project team from that evidence; the IPCC itself publishes no correlation coefficients for these pairs. Hover over any coloured cell in the matrix to see the source.

Parameter A	Parameter B	Correlation	Reason, in short
BW	MW	+0.50	Larger breeds are heavier at every age and have a larger mature weight.
BW	WG	+0.40	Well-fed animals are both heavier and gain weight faster.
Milk	Fat	-0.30	As milk yield rises, the fat percentage falls (dilution).
Milk	BW	+0.30	Larger dairy breeds and more intensive systems have heavier cows and higher yields.

Parameter A	Parameter B	Correlation	Reason, in short
Milk	DE	+0.30	More digestible feed supports higher milk yield.
DE	CP	+0.20	Within one forage, digestibility and protein fall together as the plant matures.
DE	Ym	-0.20	The IPCC assigns a lower methane conversion factor to more digestible diets.

All other pairs are zero. Four values can reasonably be changed through the manual matrix when you know your situation: BW and MW 0.8 to 0.9 when both weights come from the same survey; Milk and BW 0.10 to 0.20 for a single-breed herd; DE and CP 0.3 to 0.5 when your feed-quality uncertainty is mainly about forage maturity; DE and Ym -0.3 to -0.5 for high-grain dairy or feedlot systems.

6 Running the simulation (Tab 4: Simulate)

6.1 Settings

Before you click Run, you will have to set the following parameters.

Number of iterations. How many times the tool draws a full set of inputs and recalculates the emissions.

- 1,000: a quick test, to check that everything runs. Do not use it for reporting.
- 10,000: the minimum for reporting. The ends of the interval (the 2.5th and 97.5th percentiles) are stable.

GWP version. The global warming potentials used to convert methane and nitrous oxide into CO_2 equivalent. Choose AR4 (CH_4 = 25, N_2O = 298), AR5 (CH_4 = 28, N_2O = 265) or AR6 (CH_4 = 27, N_2O = 273; the 27 is the value for biogenic methane, which is right for cattle). The default is AR5. Use the version your reporting guidelines require.

Random seed. A number that fixes the random draws. With the same file, the same settings and the same seed, you get exactly the same results every time.

Analysis mode. Chosen on Tab 1: single year (one inventory year) or trend (several years).

Emission sources. Tick every source that applies to your inventory. A source you do not tick counts as zero. You may leave out enteric fermentation, but the enteric inputs must still be in your file, because the manure calculations use them. If enteric fermentation is not ticked, its emissions and uncertainty do not appear in the results.

6.2 Trend mode: year-to-year coefficient correlation

If you chose Trend on Tab 1, one more setting appears on Tab 4: **Year-to-year correlation**. It says how the IPCC coefficients (Cfi, Ca, Ym, Bo, MCF, EF3, EF4, EF5 and so on) move from one year to the next within one Monte Carlo draw. The activity data (N, BW, Milk and so on) are always drawn afresh for every year. This setting only concerns the per-head coefficients.

Why does it matter? Suppose your country uses the same emission factors for the whole time series, for example because the Tier 2 factors were estimated once and applied to every year. Then a draw with a high Ym in the base year should also have a high Ym in the current year. When the two years are compared, this coefficient uncertainty cancels out, and the uncertainty of the trend comes only from the changes in activity data. The IPCC (Volume 1, Chapter 3, section 3.2.3) calls this a "Type A" sensitivity. Treating the factors as independent from year to year would make the trend look much more uncertain than it is.

The three options:

- **Fully correlated coefficients (IPCC 2019 default)**: one draw of each coefficient is reused for every year. Pick this if your emission factors are IPCC defaults or come from one national study used for the whole series. This is the IPCC recommendation.

- **Partial (AR(1), rho = 0.7)**: the coefficients drift slowly from year to year (70% of last year's value, 30% fresh). Pick this if your factors are re-estimated every few years and neighbouring years share most of the same data.
- **Independent**: the coefficients are drawn afresh every year. This gives the largest trend uncertainty. Pick it only if you really re-measured every emission factor every year with new field data, which is rare for a national inventory.

A practical check: run the trend once with "Fully correlated" and once with "Independent". If the two trend uncertainties are similar, your result does not depend on this choice. If they are very different, the emission factors dominate your trend uncertainty, and your inventory report should say which assumption you used.

Trend mode runs one full simulation per year. Two counts are set on Tab 4: the iterations per year (2,000 by default, enough for the yearly chart and the trend line) and the draws for the base year and the reported year (10,000 by default, the same as a single-year run; these two years give the headline figures and the IPCC table).

6.3 Running the simulation

Click **Run Monte Carlo Simulation** (or **Run Trend Analysis** in trend mode). A progress bar shows the steps. For 10,000 iterations and one production system, a run takes 15 to 60 seconds. Inventories with many sub-categories take longer.

When the run finishes, the results open on the same tab.

To change a setting and run again, click **Back to settings** at the top of the results, change the setting, and click Run again. The new results replace the old ones. To change your input data, distributions or correlations, go back to Tab 1, 2 or 3, make the change, then return to Tab 4 and run again.

6.4 Convergence diagnostics

The diagnostics come first in the results. They tell you whether you ran enough iterations for the results to be stable. Click "Run Diagnostic" after a simulation. Four things are shown:

- **MCSE%** (Monte Carlo standard error, as a percentage of the mean): Pass below 0.5%, Warn between 0.5% and 1%, Fail at 1% or more.
- **Mean drift %**: how much the mean from the first half of the iterations differs from the second half. Pass below 2%.
- **CI drift %**: how much the ends of the 95% interval move between the two halves. Pass below 5%.
- **Convergence trace** (Figure 4.1): the running mean and the 95% interval plotted against the number of iterations. They should flatten out before the end of the run.

If a check shows Warn or Fail, go back to the settings, increase the number of iterations, and run again.

Note: each iteration is an independent draw. There are no chains, no warm-up and no burn-in, as there would be in other Monte Carlo methods.

7 Understanding results (Tab 4: Results)

Every graph in the tool has a number, a title and a caption. The number gives the tab and the position: Figure 4.2 is the second figure on Tab 4.

7.1 Headline metrics

After a run, the results open with the diagnostics (see above), then five value boxes. The masses are shown in gigagrams with three decimals by default; the choice above the boxes switches to tonnes, or shows both. The five boxes are:

- **Total CO_2eq**: the mean of the simulated total emissions.
- **Enteric CH_4**: the mean enteric fermentation emissions, with their own 95% margin of error.
- **Manure CH_4**: the mean manure methane emissions, with their margin of error.
- **Pasture N_2O**: the mean direct and indirect nitrous oxide from manure deposited on pasture, with their margin of error.
- **Total 95% MoE (%)**: the headline uncertainty. It is half the width of the 95% confidence interval, as a percentage of the mean. This is how the IPCC reports uncertainty (Volume 1, Chapter 3, Table 3.3), and it is the one uncertainty figure used everywhere in the tool.

Below the boxes, a "Headline by cattle type" card appears when your inventory has more than one cattle type (usually dairy and other). It shows one row per cattle type with the mean and the 95% margin of error for each source and for the total. With one cattle type, this card does not appear.

7.2 Uncertainty decomposition (AD vs EF)

If "Run uncertainty decomposition (AD/EF/Combined)" was ticked, the results include a bar chart (Figure 4.3). It shows the margin of error (half the 95% interval, as a percentage of the mean) in three cases:

- **AD uncertainty**: only the activity data vary (N, BW, Milk, WG). The emission factors are held at their means.
- **EF uncertainty**: only the emission factors and coefficients vary. The activity data are held at their means.
- **Combined uncertainty**: everything varies together.

If the AD uncertainty dominates, better livestock surveys would reduce the total uncertainty most. If the EF uncertainty dominates, country-specific measurements of the emission factors would help more.

7.3 By-system and by-category breakdown

- **By system**: the mean, the 95% interval, the margin of error and the coefficient of variation for each production system and each emission source. A selector lets you group the rows by cattle type (the default), by production system or by sub-category.

- **By reporting category**: the same results grouped by IPCC source category (3.A.1, 3.B.1, 3.B.5, 3.D.1.c, 3.D.2), in the layout of IPCC Table 3.3. Each row gives the mean mass of the gas itself (CH_4 or N_2O), the mean in CO_2 equivalent, the margin of error, the coefficient of variation and the 95% interval.

8 Sensitivity analysis (Tab 5)

8.1 Reading the tornado chart

The tornado chart (Figure 5.1) ranks the inputs by how much they contribute to the total uncertainty. The longest bars belong to the inputs that, if known better, would narrow the 95% interval most.

Each bar is labelled with the parameter and its sub-category, for example "Ym (cows)" and "Ym (heifers)". The sub-category tells you which animal group drives the uncertainty, so you can target your data collection at that group and not at the parameter in general. The same labels appear in the rankings table, in the Excel sheet and in the Word report. Parameters sampled per manure system carry the system in their name, for example "MCF_solid_storage (cows)".

The "Output variable" list above the chart lets you rank the inputs for one emission source only (enteric CH_4, manure CH_4, and so on). The ranking for one source covers all sub-categories.

The bars have two colours:

- **Green**: inputs whose uncertainty you can reduce with better national data (surveys, measurements).
- **Blue**: IPCC coefficients, whose uncertainty you can hardly reduce with national data.

Focus your data work on the green bars at the top of the chart.

Three downloads are available on Tab 6:

- **Excel report (.xlsx)**: the full workbook with the run settings, the IPCC summary table, the uncertainty metrics and the sensitivity results.
- **CSV**: the IPCC Table 3.3 table only, easy to import into inventory software.
- **Word report**: a narrative summary with the charts, ready to adapt for the uncertainty section of your inventory report.

9 IPCC reporting conventions

9.1 Where uncertainty results go in the national inventory report

The 2006 IPCC Guidelines (Volume 1, Chapter 3, section 3.3) describe what to report for Approach 2 (Monte Carlo):

1. the 95% confidence interval for each major source category and for the national total;
2. the results in the Table 3.3 format (the downloads on Tab 6 give it);
3. a description of the Monte Carlo approach: number of iterations, distributions used, correlation assumptions;
4. every parameter where a country-specific value replaced an IPCC default, with the justification.

The Methodology document on the Resources tab gives the complete technical description of the equations, the sampling and the convergence checks. You can cite it in your inventory report.

9.2 What IPCC expert reviewers will check

From experience with IPCC expert reviews, reviewers usually look at:

- whether the uncertainty is reported at the right level of detail (for example separately for dairy and other cattle, per emission source, when the inventory reports them separately);
- whether the Table 3.3 format and column conventions are followed;
- whether the Monte Carlo approach is described well enough to be reproduced;
- whether the number of iterations is enough for stable results;
- the basis for any country-specific uncertainty range that differs from the IPCC defaults.

10 Worked example: Country X (hypothetical dairy)

This example walks through the whole workflow with the Country X dataset.

Step 1: Load data. On Tab 1, choose "Single year". In the "Country / Example Data" list, pick "Country X (hypothetical dairy)" and confirm. The data loads (12 parameters, a smallholder dairy system). In the parameter table, note that the activity data (N, BW, Milk, WG, Fat) use normal distributions and the coefficients (Ym, DE, Bo and so on) use PERT or lognormal distributions.

Step 2: QA/QC. Go to Tab 2. Most parameters pass. Four are marked Missing: they were not in the example file and the tool filled them with IPCC defaults. A Warn on body weight would mean the value is far from the IPCC default for that animal class; a deliberate country-specific value can be documented in the inventory report rather than corrected.

Step 3: Correlations. Go to Tab 3. Select "From template (auto, time-series)". The example has a 5-year time series, so the tool computes the correlation matrix from it and shows it on the right.

Step 4: Simulate. Go to Tab 4. Set the iterations to 10,000, the GWP to AR6 and the seed to 42. Tick all emission sources. Click "Run Monte Carlo Simulation".

Step 5: Results. After about 30 seconds, the results open on the same tab. Note the 95% interval and the margin of error. Click "Run Diagnostic" to confirm that the run is stable.

Step 6: Sensitivity. Go to Tab 5. The tornado chart shows which inputs drive the uncertainty. In this example, Ym and BW for cows are usually at the top.

Step 7: Report. Go to Tab 6. Download the Excel report to see the full IPCC Table 3.3 output, or the Word report for a draft of the uncertainty section.

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